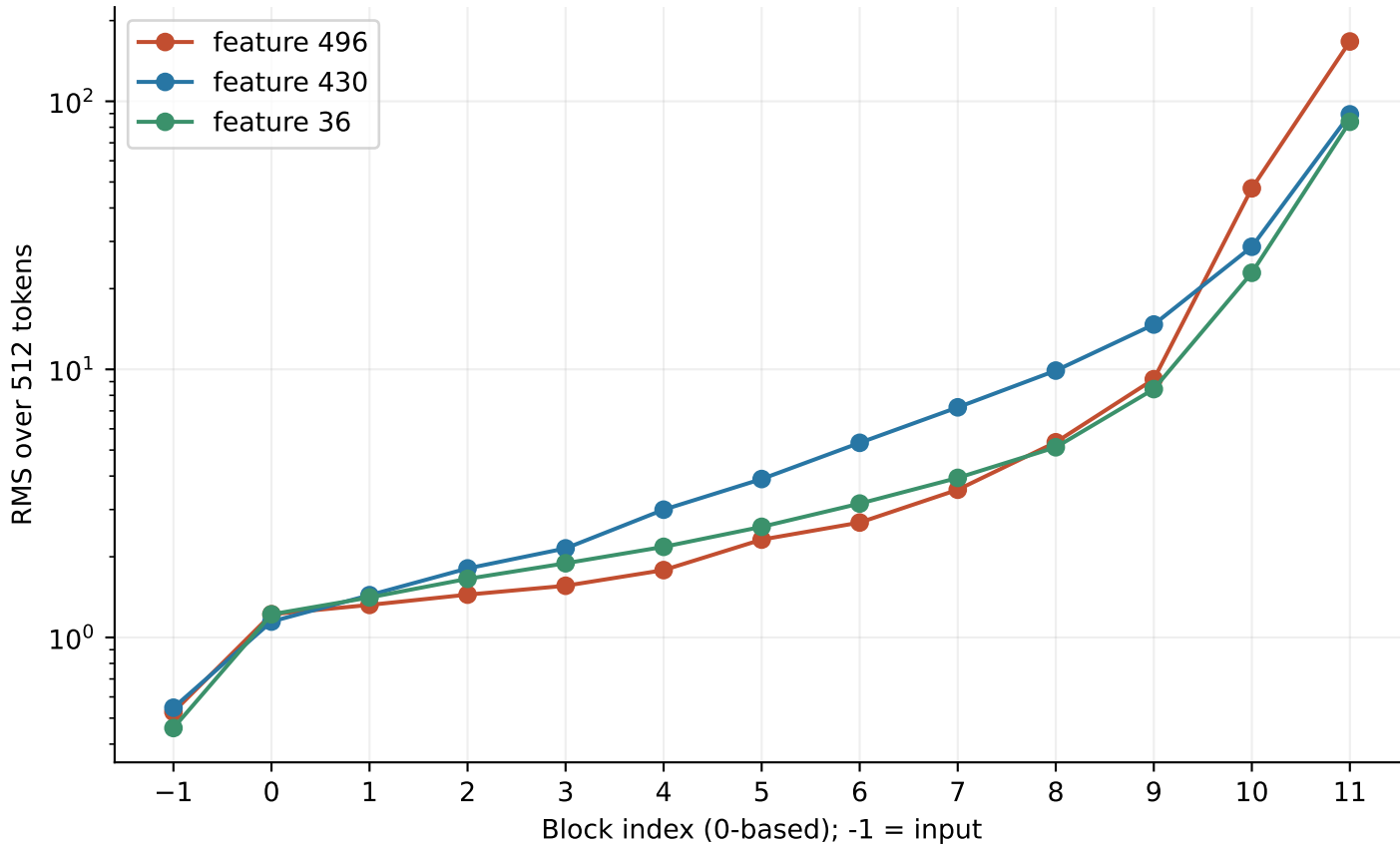


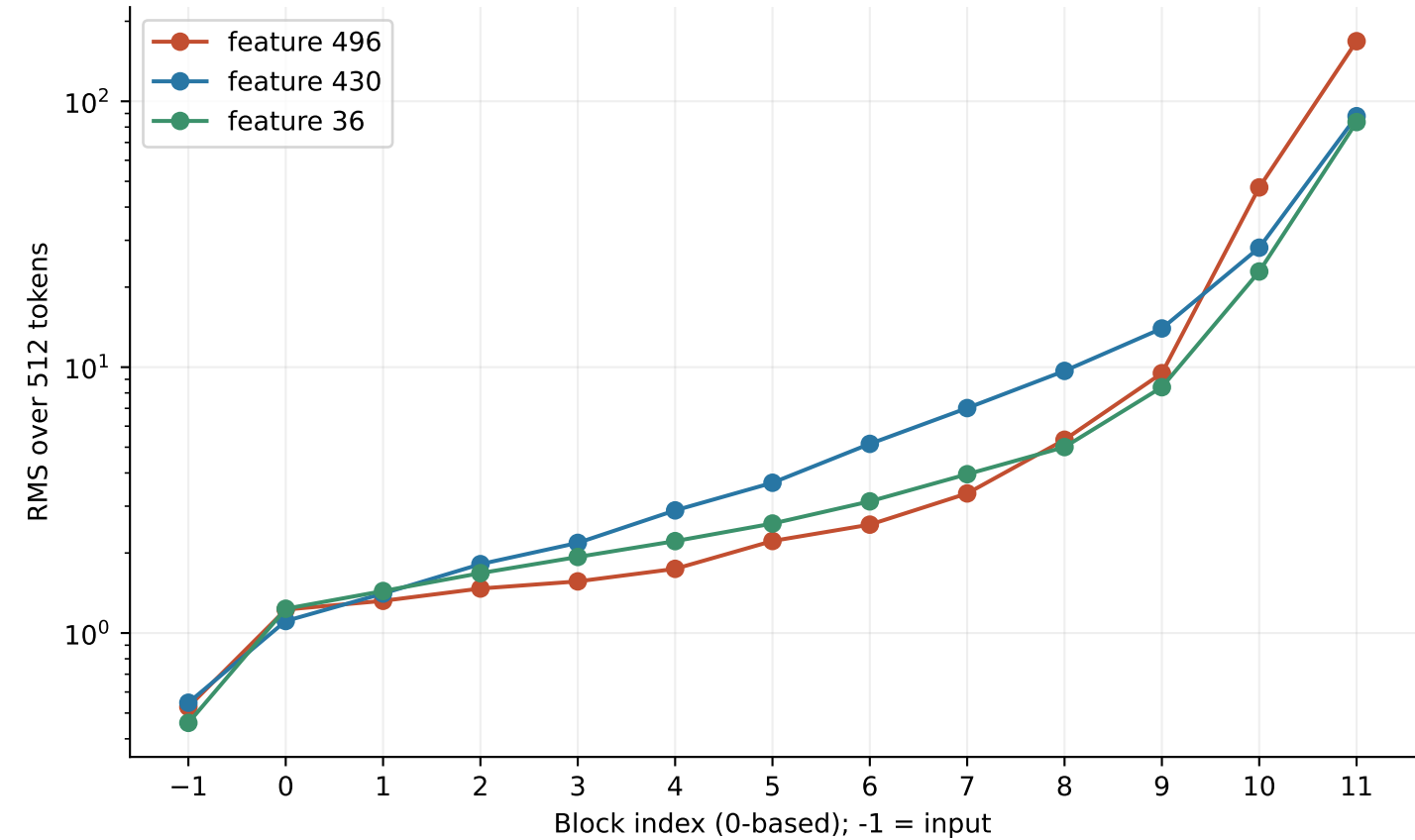
Large activations already exist before quantization

GPT-2 FP16 and Q4-body controls; Llama Q4-body reference

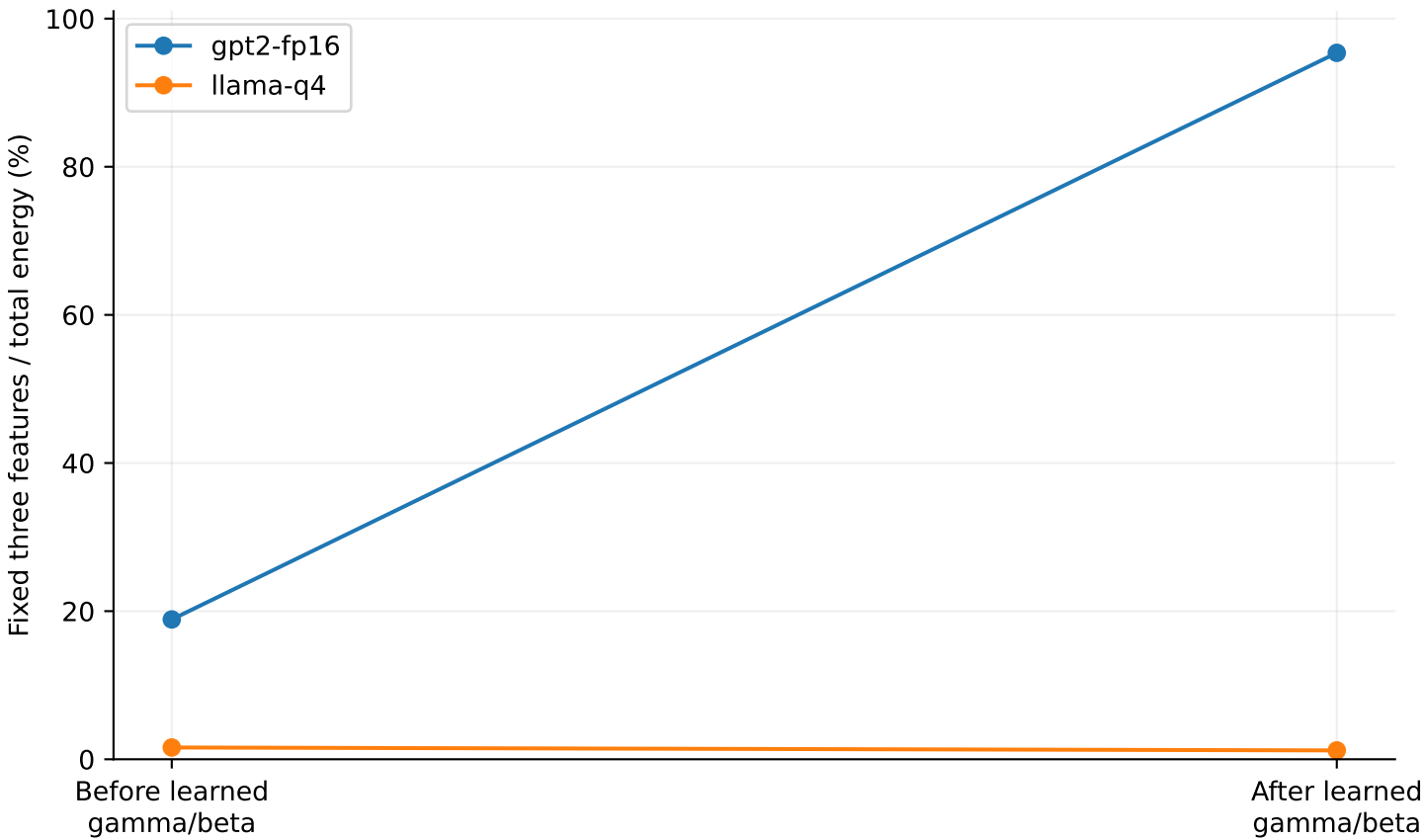
gpt2-fp16: residual-stream channel RMS



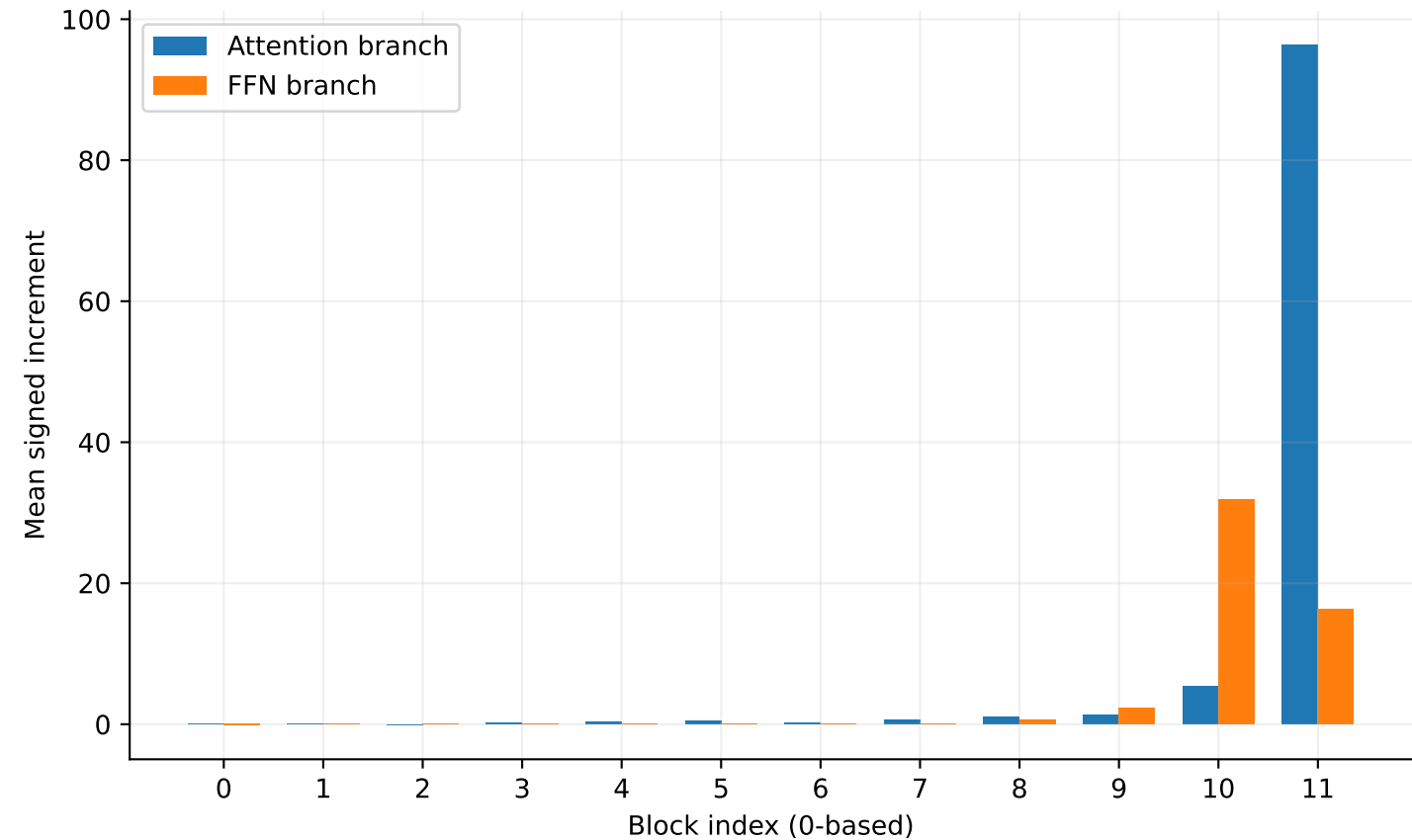
gpt2-q4: residual-stream channel RMS



Final normalization: same 512-token traces

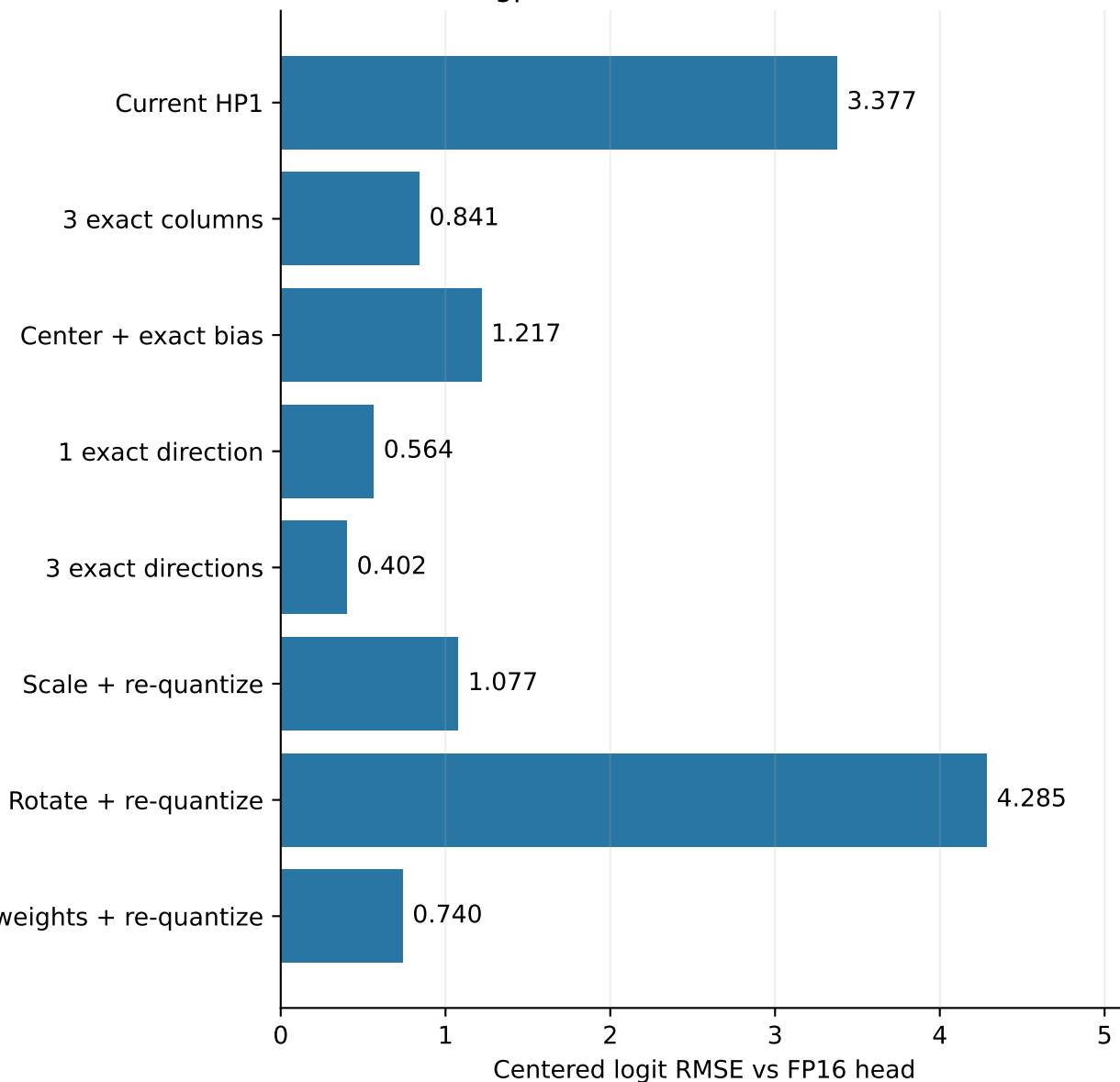


GPT-2 FP16: feature 496 mean additions



Head-only counterfactuals on held-out captured inputs
64 rows / full vocabulary; calibration = earlier 16 chunks; FP64 dot; not full PPL

gpt2: current HP1 INT4 head



llama: current HP1 INT4 head

